Acid-Base Titrations

Exam code: 0620 & 0971

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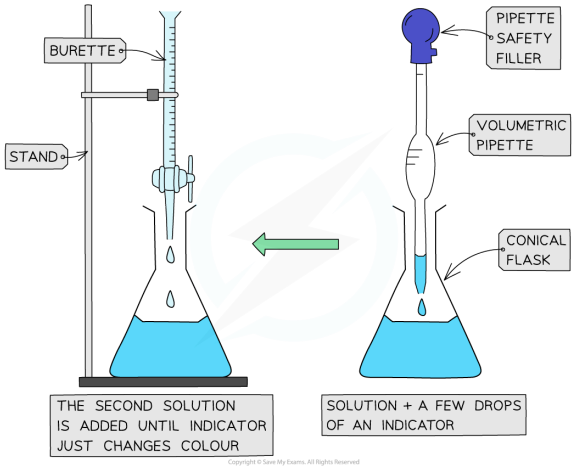
Acid-base titrations

- Titrations are a method of analysing the **concentration** of solutions
- They can determine exactly how much alkali is needed to neutralise a quantity of acid – and vice versa
- You may be asked to perform titration calculations to determine the **moles** present in a given amount or the **concentration / volume** required to **neutralise** an acid or a base
- Titrations can also be used to prepare salts

Apparatus

- 25 cm^3 volumetric pipette
- Pipette filler
- 50 cm^3 burette
- 250 cm^3 conical flask
- Small funnel
- 0.1 mol / dm^3 sodium hydroxide solution
- Sulfuric acid of unknown concentration
- A suitable indicator
- Clamp stand, clamp & white tile

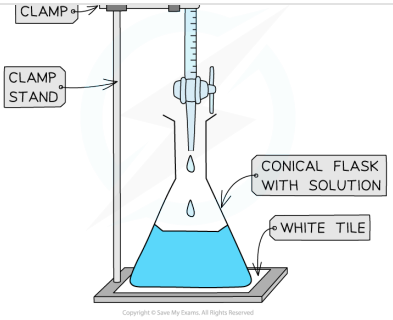


Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



The steps in performing a titration

Method

1. Use the pipette and pipette filler and place exactly 25 cm^3 sodium hydroxide solution into the conical flask
2. Using the funnel, fill the burette with hydrochloric acid placing an empty beaker underneath the tap. Run a small portion of acid through the burette to remove any air bubbles
3. Record the starting point on the burette to the nearest 0.05 cm^3
4. Place the conical flask on a white tile so the tip of the burette is inside the flask
5. Add a few drops of a suitable indicator to the solution in the conical flask
6. Perform a rough titration by taking the burette reading and running in the solution in 1 – 3 cm^3 portions, while swirling the flask vigorously
7. Quickly close the tap when the end-point is reached
   - The endpoint is when one drop causes a sharp colour change
8. Record the volume of hydrochloric acid added, in a suitable results table as shown below
   - Make sure your eye is level with the meniscus
9. Repeat the titration with a fresh batch of sodium hydroxide
10. As the rough end-point volume is approached, add the solution from the burette one drop at a time until the indicator just changes colour
11. Record the volume to the nearest 0.05 cm^3
12. Repeat until you achieve two concordant results (two results that are within 0.1 cm^3 of each other) to increase accuracy

	Rough titre	Titre 1	Titre 2	Titre 3
Final reading (cm^3)				
First reading (cm^3)				
Titre (cm^3)				

Examiner Tips and Tricks

Common errors during a titration include:

- Not removing the funnel from the burette
  - This can lead to some liquid dripping into the burette and cause false / high readings
- Not filling the jet space of the burette
  - The jet space is the part of the burette after the tap
  - Not filling this space can lead to false readings
- Reading the volume from the burette incorrectly
  - Readings should be taken from the bottom of the meniscus
  - **Careful:** The scale on the burette has 0.0 cm^3 at the top and 50 cm^3 (typically) at the bottom

Indicators

- **Indicators** are used to show the endpoint in a titration
- Wide range indicators such as litmus are not suitable for titration as they do not give a sharp colour change at the endpoint
  - However, methyl orange and phenolphthalein are very suitable
- Some of the most common indicators with their corresponding colours are shown below:

Common acid-base indicators

Indicator	Colour in acid	Colour in alkali	Colour in neutral
Litmus solution	Red	Blue	Purple
Red litmus paper	Stays red	Turns blue	No change
Blue litmus paper	Turns red	Stays blue	No change
Methyl orange	Red	Yellow	Orange
Phenolphthalein	Colourless	Pink	Colourless
Thymolphthalein	Colourless	Blue	Colourless

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